A Prime-Generated Formalization of Nagata's Factoriality Theorem in Lean 4

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Abstract

We present a Lean 4 / Mathlib formalization of Nagata's factoriality theorem for noetherian integral domains. The central hypothesis is a *prime-generated* condition on the multiplicative set: every element of the submonoid is a finite product of elements that are both prime in the ring and members of the submonoid. Under this hypothesis, if the localization is a unique factorization domain (UFD), then so is the base ring.

The formal development comprises conditional transfer lemmas for divisibility, irreducibility, and primality between a domain and a localization at a prime-generated submonoid, together with the packaging of these ingredients into a reusable Nagata theorem. As an application, we prove that if R is a noetherian UFD then the polynomial ring $R[X]$ is a UFD, by constructing a Laurent-polynomial localization and invoking the formalized theorem.

The formalization exercise surfaced a substantive mathematical issue: an earlier *prime-or-unit* hypothesis on the submonoid turned out to be degenerate, and the correct prime-generated formulation was developed as a direct consequence of the failed attempt to instantiate the restricted theorem. No public formalization of Nagata's factoriality theorem is known to us in Lean, Coq, or Isabelle.

1 Introduction

Nagata's factoriality theorem is a classical tool in commutative algebra that recovers unique factorization in a base ring from unique factorization in a suitable localization. In a standard textbook presentation [12, 14, 9, 5], the theorem states that if R is a noetherian integral domain, $S \subseteq R$ is a multiplicative set whose elements factor into primes in R, and the localization $S^{-1}R$ is a UFD, then R itself is a UFD. The argument connects localization, irreducibility, primality, and noetherianity in a way that is both algebraically instructive and technically nontrivial.

The theorem occupies a distinctive position in the landscape of formalization targets. It is short enough to be self-contained—the informal proof fits comfortably on a single page—yet it requires careful coordination of several pieces of algebraic infrastructure: localization of rings at a submonoid, transfer of divisibility across the localization map, interaction between irreducibility and primality in integral domains, and the characterization of UFDs as noetherian domains in which every irreducible is prime. It is also a gateway result: one of its immediate applications is to prove that $R[X]$ is a UFD whenever R is, via localization at powers of the indeterminate.

Motivation and scope. We describe a Lean 4 formalization, built on Mathlib [7], that packages a complete proof of Nagata's theorem together with a worked application establishing unique factorization for polynomial rings. The formalization grew out of an attempt to close a gap in Mathlib's coverage of classical factoriality results: while the library provides extensive infrastructure for unique factorization, localization, and polynomial algebra, no Nagata-style transfer theorem was available.

An earlier version of this project stated the main result under the hypothesis that every element of the multiplicative set S is prime or a unit—a condition that is superficially clean but essentially degenerate for any submonoid with more than one non-unit prime generator. The present development replaces that restricted statement with a *prime-generated* hypothesis and rebuilds the supporting machinery around the corrected formulation. We discuss the mathematical reasons in Section 2 and the way the formalization surfaced the problem in Section 8.

Contributions. The formal contribution is threefold:

1. A formalization of Nagata's factoriality theorem under a prime-generated hypothesis on the multiplicative set (§3).
2. Conditional transfer lemmas for divisibility, irreducibility, and primality between a noetherian domain and its localization (§4).
3. A theorem-driven application: if R is a noetherian UFD then $R[X]$ is a UFD, proved via a Laurent-polynomial localization and the formalized Nagata theorem (§7).

Beyond these formal artifacts, the project contributes a case study in the interaction between formalization and mathematical hypothesis design: the failure of the initial prime-or-unit formulation was a genuine mathematical deficiency that the formal setting made visible earlier and more sharply than a textbook exposition might.

Outline. Section 2 reviews the mathematical background. Section 3 gives the formal statement. Section 4 describes the proof architecture. Section 6 discusses file organization and API design. Section 7 presents the polynomial-ring application. Section 8 draws proof-engineering and mathematical lessons. Section 9 surveys related work. Section 11 describes the artifact. Section 12 concludes.

2 Mathematical background

We briefly recall the algebraic ingredients that enter the Nagata argument, with an emphasis on the points most relevant to formalization.

2.1 Localization at a multiplicative submonoid

Let R be a commutative ring and $S \subseteq R$ a multiplicative submonoid (i.e., $1 \in S$ and S is closed under multiplication). The *localization* of R at S, written $S^{-1}R$, is the ring of fractions a/s with $a \in R$ and $s \in S$, subject to the equivalence relation $a/s = b/t$ iff there exists $u \in S$ with $u(at - bs) = 0$. When R is an integral domain and $0 \notin S$, the relation simplifies to $at = bs$.

The ring homomorphism $\iota\colon R \to S^{-1}R$ sending $a \mapsto a/1$ is injective precisely when S consists of non-zero-divisors, which is automatic for a domain with $0 \notin S$. Every $s \in S$ becomes a unit in $S^{-1}R$ since $(s/1) \cdot (1/s) = 1$. This is the key property that the Nagata argument exploits: denominators from S can be "cleared" in $S^{-1}R$.

2.2 Unique factorization domains

A commutative integral domain R is a *unique factorization domain* (UFD) if every nonzero non-unit element can be written as a finite product of irreducible elements, and this factorization is essentially unique (up to reordering and multiplication by units). An equivalent characterization, which is the one used in our formalization, is:

Proposition 2.1. *A commutative integral domain R is a UFD if and only if:*

1. *R is a well-founded divisibility monoid (i.e., every element admits a factorization into irreducibles), and*
2. *every irreducible element of R is prime.*

Condition (1) is automatic in any noetherian domain, since the ascending chain condition prevents infinite chains of proper divisors. The Nagata argument therefore reduces to establishing condition (2): every irreducible is prime.

2.3 The classical Nagata argument

Let R be a noetherian integral domain, $S \subseteq R$ a submonoid whose elements factor into primes, and suppose $S^{-1}R$ is a UFD. We must show that every irreducible $p \in R$ is prime. Two cases arise.

Case 1: p divides some $s \in S$. Because s factors as a product of primes belonging to S and p is irreducible, p must be associate to one of those prime factors, hence prime.

Case 2: p does not divide any element of S. Then the image $p/1$ in $S^{-1}R$ is not a unit and remains irreducible. Since $S^{-1}R$ is a UFD, $p/1$ is prime. One then pulls primality back: if $p \mid ab$ in R, then $p/1 \mid (ab)/1$ in $S^{-1}R$, so $p/1 \mid a/1$ or $p/1 \mid b/1$. Lifting this divisibility to R (using avoidance to clear denominators) gives $p \mid a$ or $p \mid b$.

Each step involves nontrivial interaction with the localization machinery. It is the careful disentangling of these interactions into modular, reusable lemmas that constitutes the main formal engineering contribution.

2.4 Prime-generated vs. prime-or-unit

The hypothesis that "every element of S factors as a product of primes belonging to S" is sometimes stated more simply: "every element of S is prime or a unit." This simpler form suffices for the localization $S = \{1, p, p^2, \ldots\}$ generated by a single prime p, and appears in some textbook presentations that only need this case.

However, for a general submonoid the prime-or-unit condition is far too restrictive. If p and q are two distinct non-unit primes in S, then $pq \in S$ (since S is a submonoid) but pq is neither prime nor a unit. Thus the prime-or-unit condition can hold only when S is generated by at most one prime up to associates together with the group of units—a severe restriction that excludes most interesting applications.

The *prime-generated* condition avoids this collapse by allowing products: it asks that every $s \in S$ factor as a finite product of elements that are individually prime and individually members of S. In a localization argument, this is exactly what one needs: to clear a denominator $s \in S$ from a fraction, one factors $s = q_1 \cdots q_n$ into primes, and because each q_i becomes a unit in $S^{-1}R$, so does s.

Remark 2.2. The prime-generated condition is formalized using Lean's `Multiset` type to represent the finite product, avoiding ordering issues that would arise with lists. The formal predicate is:

$$\mathit{PrimeGenerated}(S) \stackrel{\text{def}}{=} \forall\, s \in S,\ \exists\, f : \mathrm{Multiset}(R),\ \big(\forall\, q \in f,\ q \in S \wedge \mathrm{Prime}(q)\big)\ \wedge\ \prod f = s.$$

3 Formal statement

We work over a commutative ring R that is both an integral domain and noetherian.

Definition 3.1 (Prime-generated submonoid)**.** A submonoid $S \subseteq R$ is *prime-generated* if every $s \in S$ can be written as a finite product $s = q_1 \cdots q_n$ where each q_i satisfies $q_i \in S$ and $\mathrm{Prime}(q_i)$ in R. The empty product yields $s = 1$, covering the unit element of the submonoid.

Definition 3.2 (Avoidance)**.** An element $p \in R$ *avoids* a submonoid S if p does not divide any element of S:

$$\mathit{Avoids}(S, p) \stackrel{\text{def}}{=} \forall\, s \in S,\ \neg\,(p \mid s).$$

The avoidance predicate captures the condition under which the image of p in $S^{-1}R$ cannot be a unit. It is a hypothesis of several transfer lemmas and arises naturally from the case analysis in the key lemma.

Theorem 3.3 (Nagata)**.** *Let R be a noetherian integral domain and $S \subseteq R$ a prime-generated submonoid. If the localization $S^{-1}R$ is a UFD, then R is a UFD.*

The formal statement appears as `nagata_theorem` in the file `Nagata/Theorem.lean`. Its proof invokes two ingredients: the fact that a noetherian domain is a well-founded divisibility monoid (giving condition (1) of Proposition 2.1), and the key lemma (giving condition (2)).

The prime-generator packaging. The repository also packages a convenient family of corollaries. The lemma `primeGenerated_closure_of_primes` shows that if $\mathcal{P}$ is any set of prime elements of R, then the submonoid closure($\mathcal{P}$) is automatically prime-generated. As a consequence, `Nagata/Theorem.lean` exports two additional entry points:

- `nagata_theorem_of_prime_generators` specializes Theorem 3.3 to localizations at submonoids generated by arbitrary sets of primes.
- `nagata_theorem_of_finite_prime_generators` covers the especially common finite-generator case.

These give the repository a broader mathematical surface than a single isolated theorem statement.

Remark 3.4 (The prime-or-unit variant)*.* The earlier prime-or-unit variant is retained as `nagata_theorem_of_prime_or_unit` in the same file. It is proved *separately*, with its own chain of transfer lemmas that work under the hypothesis $\forall\, s \in S,\ \mathrm{Prime}(s) \vee \mathrm{IsUnit}(s)$. In particular, it is *not* derived as a corollary of the prime-generated theorem; the two proof lines are independent.

This design reflects the fact that the prime-or-unit transfer lemmas are simpler (they do not require multiset induction) and may be useful in their own right for the single-generator case. Retaining both chains also means that the compatibility theorem can be checked without depending on the more complex prime-generated infrastructure.

4 Proof architecture

The proof proceeds in four layers, each providing reusable components for the next. We describe these layers bottom-up, giving both the mathematical content and the names of the corresponding Lean declarations.

4.1 Layer 1: prime-generated submonoid API

The file `Nagata/Lemmas.lean` introduces the `PrimeGenerated` and `Avoids` predicates (Definitions 3.1 and 3.2). The basic consequences established at this layer are:

`zero_notMem_of_primeGenerated`.
A prime-generated submonoid does not contain zero. If $0 \in S$ then 0 would factor as a product of primes, but a product of primes is nonzero.

`primeGenerated_powers`.
If p is prime, then the submonoid $\{1, p, p^2, \ldots\}$ is prime-generated. The factorization of p^n is simply n copies of p.

`primeGenerated_closure_of_primes`.
If every element of a set $\mathcal{P}$ is prime, then the submonoid closure of $\mathcal{P}$ is prime-generated. This is the packaging lemma that turns the main theorem into a ready-made criterion for localizations at submonoids generated by prime families.

`multiset_prod_mem_of_factors`.
If every element of a multiset belongs to a submonoid S, then so does the product.

These results are straightforward but foundational. In particular, the zero-exclusion lemma is needed to establish that $S^{-1}R$ is an integral domain, since Mathlib requires $0 \notin S$ as a `Fact` instance for the domain structure on the localization.

4.2 Layer 2: conditional transfer lemmas

The bulk of the formal development consists of lemmas that transfer divisibility and factorization properties between R and $S^{-1}R$. Each lemma relies on a specific combination of the prime-generated hypothesis and an avoidance hypothesis.

Lemma 4.1 (Irreducibles dividing prime factorizations are prime)**.** *Let f be a multiset of prime elements of R, and let $p \in R$ be irreducible. If $p \mid \prod f$, then p is prime.*

Proof sketch. By induction on f. If f is empty then $p \mid 1$, contradicting irreducibility. If $f = q :: f'$, then $q \mid p \cdot d$ for some d, and primality of q gives $q \mid p$ or $q \mid d$. In the first case, q and p are associates, so p is prime. In the second case, cancel q and apply the induction hypothesis. □

The Lean name is `prime_of_irreducible_of_dvd_prime_factors`. A direct consequence is:

Lemma 4.2. *If S is prime-generated, p is irreducible, $s \in S$, and $p \mid s$, then p is prime.*

(Lean name: `prime_of_irreducible_of_dvd_mem_primeGenerated`.) This combines the previous lemma with the prime factorization of s supplied by the prime-generated hypothesis.

Lemma 4.3 (Lifting divisibility from the localization)**.** *Let S be prime-generated, p irreducible, and suppose p avoids S. If $p/1 \mid a/1$ in $S^{-1}R$, then $p \mid a$ in R.*

Proof sketch. From the divisibility in the localization one obtains $s \in S$ and $c \in R$ with $s \cdot a = p \cdot c$. Factor $s = q_1 \cdots q_n$ into primes from S. Because p avoids S, we have $p \nmid q_i$ for each i. An auxiliary result then peels off the prime factors one at a time: at each step, the prime q_i divides $p \cdot$(something) but cannot divide p, so by primality of q_i it divides the cofactor. After all factors are consumed, $p \mid a$. □

Lean name: `dvd_of_localization_dvd_primeGenerated`.

Lemma 4.4 (Transfer of irreducibility)**.** *Let S be prime-generated, p irreducible, and suppose p avoids S. Then $p/1$ is irreducible in $S^{-1}R$.*

Proof sketch. First, $p/1$ is not a unit: otherwise $p/1 \mid 1/1$, which would lift to p dividing some element of S, contradicting avoidance. Second, if $p/1 = (a/s)(b/t)$ in $S^{-1}R$, one clears denominators to get $p \cdot (st) = ab$ in R. The product $st \in S$ factors as a multiset of primes; the auxiliary lemma `split_prime_factors_of_mul_eq` partitions these factors between the two sides, reducing the equation to $p = a'b'$ modulo units from S. Irreducibility of p forces one factor to be a unit. □

Lean name: `localization_irreducible_of_irreducible_primeGenerated`. The auxiliary `split_prime_factors_of_mul_eq` is the longest single lemma in the development: it partitions a multiset of prime factors across the two sides of a product equation by induction on the multiset, using primality of each factor to decide which side absorbs it.

Lemma 4.5 (Transfer of primality)**.** *Let S be prime-generated, p irreducible, p avoids S, and $p/1$ prime in $S^{-1}R$. Then p is prime in R.*

Proof sketch. If $p \mid ab$ in R, then $p/1 \mid (ab)/1$ in $S^{-1}R$. Primality of $p/1$ gives $p/1 \mid a/1$ or $p/1 \mid b/1$. Lemma 4.3 lifts the divisibility back to R. □

Lean name: `prime_of_localization_prime_primeGenerated`.

The denominator-clearing helper. Underlying both the divisibility lift and the irreducibility transfer is an auxiliary lemma that merits explicit mention:

Lemma 4.6 (`dvd_of_mul_eq_prime_factors`)**.** *Let f be a multiset of prime elements of R, let p be irreducible, and suppose p does not divide any element of f. If $(\prod f) \cdot a = p \cdot c$, then $p \mid a$.*

The proof proceeds by induction on f. In the base case the equation $1 \cdot a = p \cdot c$ gives $p \mid a$ directly. In the inductive step, the head prime q divides $p \cdot c$; since q is prime and $q \nmid p$ (by hypothesis), we have $q \mid c$, say $c = q \cdot c'$. Canceling q from both sides yields $(\prod f') \cdot a = p \cdot c'$, and the induction hypothesis applies.

This lemma is invoked whenever a denominator from S must be "cleared" from a divisibility relation in R. Its correctness depends critically on the fact that each factor of the denominator is prime (so that primality can redirect the divisibility) and that the irreducible p does not divide any of those factors (the avoidance hypothesis at work one factor at a time).

Comparing the two proof chains. As noted in Section 3, the development maintains two independent chains of transfer lemmas: one for the prime-generated hypothesis, one for the prime-or-unit hypothesis. Table 1 summarizes the correspondence.

Table 1: Correspondence between the two chains of transfer lemmas. Each row shows a pair of lemmas that prove the same mathematical statement under different hypotheses on the submonoid S.

Prime-generated chain	**Prime-or-unit chain**
`dvd_of_localization_dvd_primeGenerated`	`dvd_of_localization_dvd`
`localization_irreducible_of_irreducible_primeGenerated`	`localization_irreducible_of_irreducible`
`prime_of_localization_prime_primeGenerated`	`prime_of_localization_prime`
`nagata_key_lemma_primeGenerated`	`nagata_key_lemma`

The prime-or-unit chain avoids multiset induction entirely: because every element of S is individually prime or a unit, there is no need to factor a denominator into multiple primes. Instead, the proofs use a direct case-by-case argument on each denominator. This makes the prime-or-unit proofs shorter and more elementary, but they apply only to submonoids generated by at most one prime up to associates.

The dual-chain design was a deliberate engineering choice. The prime-or-unit proofs served as a "warm-up" implementation that validated the overall proof strategy before the more complex multiset-inductive arguments were attempted. Retaining both chains in the final artifact also provides a pedagogical benefit: a reader can study the simpler chain first to understand the structure, then see how each step generalizes when prime factorizations of denominators are needed.

4.3 Layer 3: the key lemma

Lemma 4.7. *Let S be a prime-generated submonoid of a noetherian integral domain R. If $S^{-1}R$ is a UFD, then every irreducible element of R is prime.*

Lean name: `nagata_key_lemma_primeGenerated`. The proof combines the transfer lemmas via a case split on an irreducible element $p \in R$:

1. If p divides some $s \in S$, then p is prime by Lemma 4.2.
2. If p avoids S, then $p/1$ is irreducible in $S^{-1}R$ by Lemma 4.4. Since $S^{-1}R$ is a UFD, every irreducible in $S^{-1}R$ is prime, so $p/1$ is prime. Then p is prime by Lemma 4.5.

This case split is the structural heart of the Nagata argument. Every other formal result in the development serves to make one of these two branches go through.

4.4 Layer 4: final packaging

The public theorem `nagata_theorem` (Theorem 3.3) combines the key lemma with the standard UFD characterization (Proposition 2.1):

1. Condition (1)—every element factors into irreducibles—holds because R is noetherian (via a `WfDvdMonoid` instance).
2. Condition (2)—every irreducible is prime—is the key lemma.

The packaging lemma `ufd_of_factorization_and_primes` (in `Basic/UFD.lean`) assembles these two conditions into a `UniqueFactorizationMonoid` instance. The user supplies a noetherian domain, a prime-generated submonoid, and a UFD instance on the localization, and receives a UFD instance on the base ring.

UFD characterization and uniqueness. The file `Basic/UFD.lean` provides the foundational interface between our development and Mathlib's `UniqueFactorizationMonoid` class. The key result is the biconditional characterization:

$$R \text{ is a UFD} \iff R \text{ is a } \texttt{WfDvdMonoid} \wedge \forall p \in R,\ \text{Irreducible}(p) \Rightarrow \text{Prime}(p).$$

This is formalized as:

`ufd_iff_factorization_and_irreducibles_prime`

As a consequence, the reverse implication—that irreducible implies prime in a UFD—is recorded as `prime_iff_irreducible_of_ufd`, completing the loop that the Nagata argument exploits in Case 2 of the key lemma.

The file also includes explicit proofs of factorization uniqueness. The theorem `factorization_unique` shows that two factorizations of the same element into irreducibles are related by a bijection that pairs associate elements. This result is not needed for the Nagata argument itself—it only establishes the *existence* of the UFD structure—but it serves two purposes: it validates the correctness of the UFD characterization, and it provides a standalone formal proof of uniqueness that can be cited independently.

5 Formal Lean statements

Following the convention of recent formalization papers (e.g., [3, 13, 6]), we include the Lean 4 source text for the central definitions and theorems. All declarations shown below are kernel-checked; the full machine-readable development is available in the accompanying artifact.

5.1 Core predicates

The two predicates that underpin the entire development are defined in `Nagata/Lemmas.lean`:

```
def PrimeGenerated {$\alpha$ : Type*} [CommRing $\alpha$] (S : Submonoid $\alpha$) :
    Prop :=
  $\forall$ s : $\alpha$, s $\in$ S $\to$ $\exists$ f : Multiset $\alpha$, ($\forall$
     q $\in$ f, q $\in$ S $\wedge$ Prime q) $\wedge$ f.prod = s

def Avoids {$\alpha$ : Type*} [CommRing $\alpha$] [IsDomain $\alpha$]
    (S : Submonoid $\alpha$) (p : $\alpha$) : Prop :=
  $\forall$ s : $\alpha$, s $\in$ S $\to$ $\neg$ p $\mid$ s
```

The `PrimeGenerated` predicate uses `Multiset` rather than `List` to represent the factorization, eliminating any dependence on the ordering of factors. The `Avoids` predicate is stated as a universal negation over submonoid membership, which integrates naturally with Lean's `Classical.em` for the case split in the key lemma.

5.2 The main theorem and its variants

The public theorem surface resides in `Nagata/Theorem.lean`:

```
theorem nagata_theorem
    {R : Type*} [CommRing R] [IsDomain R] [IsNoetherianRing R]
    (S : Submonoid R) (hS : PrimeGenerated S)
    (hUFD : @UniqueFactorizationMonoid (Localization S) $\langle\cdots\rangle$) :
    UniqueFactorizationMonoid R
```

The explicit `@UniqueFactorizationMonoid (Localization S) ...` is necessary because Lean must be told which `CommMonoidWithZero` instance to use for the localization. The instance is constructed in a `by` block that first registers $0 \notin S$ as a `Fact` and then invokes typeclass inference. We elide this block as "$\langle\cdots\rangle$" in the listings below for readability; the full text is in the artifact.

The prime-generator packaging family:

```
theorem nagata_theorem_of_prime_generators
    {R : Type*} [CommRing R] [IsDomain R] [IsNoetherianRing R]
    (s : Set R) (hs : $\forall$ q $\in$ s, Prime q)
    (hUFD : @UniqueFactorizationMonoid
              (Localization (Submonoid.closure s)) $\langle\cdots\rangle$) :
    UniqueFactorizationMonoid R
```

```
theorem nagata_theorem_of_finite_prime_generators
    {R : Type*} [CommRing R] [IsDomain R] [IsNoetherianRing R]
    (s : Finset R) (hs : $\forall$ q $\in$ s, Prime q)
    (hUFD : @UniqueFactorizationMonoid
              (Localization (Submonoid.closure ($\uparrow$s : Set R))) $\langle\cdots
    \rangle$) :
    UniqueFactorizationMonoid R
```

These entry points relieve the user from constructing the `PrimeGenerated` witness manually: the intermediate lemma `primeGenerated_closure_of_primes` handles the closure argument internally.

5.3 Key transfer lemmas

The transfer lemmas that constitute the technical core (Layer 2 of the proof architecture) have the following types:

```
-- Lifting divisibility from the localization
theorem dvd_of_localization_dvd_primeGenerated
    {$\alpha$ : Type*} [CommRing $\alpha$] [IsDomain $\alpha$]
    {S : Submonoid $\alpha$} (hS : PrimeGenerated S)
    {p a : $\alpha$} (hp : Irreducible p) (havoid : Avoids S p)
    (hdiv : Localization.of p $\mid$ Localization.of a) : p $\mid$ a

-- Transfer of irreducibility
theorem localization_irreducible_of_irreducible_primeGenerated
    {$\alpha$ : Type*} [CommRing $\alpha$] [IsDomain $\alpha$]
    {S : Submonoid $\alpha$} (hS : PrimeGenerated S)
    {p : $\alpha$} (hp : Irreducible p) (havoid : Avoids S p) :
    Irreducible (Localization.of p)

-- Transfer of primality
theorem prime_of_localization_prime_primeGenerated
    {$\alpha$ : Type*} [CommRing $\alpha$] [IsDomain $\alpha$]
    {S : Submonoid $\alpha$} (hS : PrimeGenerated S)
    {p : $\alpha$} (hp : Irreducible p) (havoid : Avoids S p)
    (hploc : Prime (Localization.of p)) : Prime p
```

Note the uniform pattern: each transfer lemma takes the `PrimeGenerated` hypothesis, an `Irreducible` witness, and the `Avoids` hypothesis, then produces a factorization-theoretic conclusion. The modularity of this interface is what makes the key lemma's case split clean.

5.4 The application chain

The application theorems in `Applications/Laurent.lean` read:

```
theorem laurentPolynomial_uniqueFactorizationMonoid
    {R : Type*} [CommRing R] [IsDomain R]
    [UniqueFactorizationMonoid R] :
    UniqueFactorizationMonoid R[T;T$^{-1}$]

theorem polynomial_uniqueFactorizationMonoid_of_laurent
    {R : Type*} [CommRing R] [IsDomain R]
    [IsNoetherianRing R] [UniqueFactorizationMonoid R[T;T$^{-1}$]] :
    UniqueFactorizationMonoid R[X]
```

```
theorem polynomial_uniqueFactorizationMonoid_via_nagata
    {R : Type*} [CommRing R] [IsDomain R]
    [IsNoetherianRing R] [UniqueFactorizationMonoid R] :
    UniqueFactorizationMonoid R[X]
```

The final theorem is the composite: from a noetherian UFD R, it first constructs the Laurent UFD instance and then descends via the Nagata theorem to obtain a UFD instance on $R[X]$. Lean's typeclass resolution automatically chains through the intermediate instances, so the user-facing API is a single theorem with three typeclass hypotheses and one conclusion.

6 File organization and API design

The Lean development is organized into four top-level modules. Table 2 summarizes the structure.

Table 2: File structure of the formalization.

Module	**File**	**Contents**
`Basic/`	`Ring.lean`	Mathlib imports for ring infrastructure
	`Divisibility.lean`	Divisibility, associatedness, prime–irreducible bridge
	`Noetherian.lean`	`WfDvdMonoid` from noetherianity
	`UFD.lean`	UFD characterization
`Localization/`	`MultSet.lean`	Submonoid utilities (zero exclusion)
	`Localization.lean`	Localization wrapper API
	`Properties.lean`	Re-export hub
`Nagata/`	`Lemmas.lean`	`PrimeGenerated`, `Avoids`, all transfer lemmas
	`Theorem.lean`	`nagata_theorem` and variants
`Applications/`	`Laurent.lean`	Laurent-polynomial application
	`Gauss.lean`	Gauss-lemma wrappers
	`Examples.lean`	Thin wrappers around Mathlib instances

6.1 Design principles

Concrete localization. Mathlib provides two views of localization: the concrete quotient type `Localization S` and the abstract typeclass `IsLocalization`. Our development works primarily with the concrete type, wrapping it in a thin API that hides the underlying quotient construction. The Nagata argument requires explicit manipulation of numerators and denominators, and the concrete type makes these operations definitionally transparent. The tradeoff is that the results are stated for `Localization S` rather than for an arbitrary `IsLocalization` instance, which would require an additional abstraction step for Mathlib integration.

Fact-based zero exclusion. The condition $0 \notin S$ is carried as a `Fact` instance rather than as an explicit hypothesis on each lemma. This reduces hypothesis clutter and allows Lean's typeclass machinery to supply the proof automatically once established from the prime-generated hypothesis.

Dual proof chains. The file `Nagata/Lemmas.lean` contains two parallel chains of transfer lemmas: one under the prime-generated hypothesis and one under the prime-or-unit hypothesis. The prime-or-unit chain is simpler—it does not require multiset induction—and serves as a self-contained proof for the single-generator case. The two chains share the `Avoids` predicate and localization API but are otherwise independent.

Separation of concerns. The `Basic/` module re-proves or wraps several Mathlib results as named lemmas in the project namespace. While redundant from a library perspective, these wrappers stabilize the interface against Mathlib API changes and make the proof scripts more readable by giving short, descriptive names to frequently used facts.

6.2 The localization wrapper API

The file `Localization/Localization.lean` provides a thin API layer over Mathlib's localization, exposing four primitive operations:

`Localization.mk (a : R) (s : S).`
The raw constructor, producing the fraction a/s.

`Localization.of (a : R).`
The canonical ring homomorphism $\iota\colon R \to S^{-1}R$, sending $a \mapsto a/1$.

`Localization.surj (z : S⁻¹R).`
Every element of the localization can be written as a/s for some $a \in R$ and $s \in S$. This is the formal version of the "clearing denominators" step used throughout the Nagata argument.

`Localization.mk_eq_iff.`
Equality in the localization: $a/s = b/t$ if and only if $at = bs$ (using the domain hypothesis to simplify the general equivalence relation).

Two further results in the same file play a structural role:

- `algebraMap_injective`: the canonical map ι is injective when $0 \notin S$, which ensures that R embeds faithfully into $S^{-1}R$.
- `dvd_of_iff`: the divisibility characterization $\iota(a) \mid \iota(b) \Leftrightarrow \exists\, s \in S,\ a \mid s \cdot b$, which is the main computational interface between the localization and the ring-side arguments.

The zero-exclusion infrastructure resides in `Localization/MultSet.lean`, which establishes that if every element of S is prime or a unit (or, more generally, if S is prime-generated), then $0 \notin S$, every member of S is a non-zero-divisor, and S embeds into the submonoid of non-zero-divisors. These results are registered as `Fact` instances so that downstream lemmas can access the domain structure on $S^{-1}R$ without threading the hypothesis explicitly.

6.3 Dependency structure and size

Table 3 gives a per-file breakdown of the formalization. The dominant module is `Nagata/Lemmas.lean`, which accounts for roughly half the total code. This is expected: the transfer lemmas involve the most intricate case analysis and multiset induction in the project.

The dependency graph is strictly layered: `Basic/` depends only on Mathlib; `Localization/` depends on `Basic/` and Mathlib; `Nagata/` depends on `Localization/`; and `Applications/` depends on `Nagata/`. No circular dependencies exist, and the layering ensures that each module can be type-checked independently once its predecessors are built.

Table 3: Per-file line counts (non-blank Lean source lines).

File	Lines	File	Lines
`Nagata/Lemmas.lean`	489	`Applications/Laurent.lean`	98
`Localization/Localization.lean`	140	`Basic/UFD.lean`	65
`Basic/Divisibility.lean`	54	`Nagata/Theorem.lean`	54
`Localization/MultSet.lean`	22	`Applications/Gauss.lean`	15
`Basic/Noetherian.lean`	11	`Applications/Examples.lean`	16

7 Application: polynomial rings

The application file `Applications/Laurent.lean` puts the formalized theorem to work, proving a textbook result by a route that genuinely depends on the Nagata machinery.

Corollary 7.1. *If R is a noetherian UFD, then the polynomial ring $R[X]$ is a UFD.*

The proof assembles five formal ingredients.

7.1 Step 1: the submonoid is prime-generated

Let $S = \{1, X, X^2, \ldots\} \subseteq R[X]$ be the submonoid of powers of the indeterminate. The element X is prime in $R[X]$ (formalized as a wrapper around Mathlib's `Polynomial.prime_X`). By `primeGenerated_powers`, the submonoid of powers of a prime element is prime-generated.

7.2 Step 2: the localization is the Laurent polynomial ring

Mathlib provides the fact that the Laurent polynomial ring $R[X, X^{-1}]$ is a localization of $R[X]$ at the powers of X. This is recorded as a wrapper around `LaurentPolynomial.isLocalization`.

7.3 Step 3: the Laurent polynomial ring is a UFD

The theorem `laurentPolynomial_uniqueFactorizationMonoid` proves that if R is a UFD, then $R[X, X^{-1}]$ is a UFD. This is the most substantial component of the application and does not follow from a Mathlib instance.

The proof works with the Mathlib-provided UFD structure on $R[X]$ and transfers it to $R[X, X^{-1}]$ by showing that every nonzero element of the Laurent ring has a prime factorization. Given $f \in R[X, X^{-1}]$ with $f \neq 0$, one writes $f \cdot X^n = p$ for some polynomial $p \in R[X]$ and exponent n, using the Mathlib lemma `LaurentPolynomial.exists_T_pow`. One then separates the X-adic part of p from its X-free part: $p = X^m \cdot q$ where $X \nmid q$.

The polynomial q admits a prime factorization in $R[X]$; each prime factor r_i satisfies $r_i \nmid X$. An auxiliary lemma shows that if a polynomial r is prime in $R[X]$ and does not divide X, then its image in $R[X, X^{-1}]$ is prime. This uses the ideal-theoretic criterion: it shows that the image of the principal ideal (r) under the localization map remains prime, via Mathlib's `IsLocalization.isPrime_of_isPrime_disjoint` and the disjointness of (r) from the powers of X.

Applying this to each prime factor of q produces a prime factorization of f in $R[X, X^{-1}]$ (after absorbing the powers of X as units).

7.4 Step 4: descending from the localization

The theorem `polynomial_uniqueFactorizationMonoid_of_laurent` shows that if R is noetherian and $R[X, X^{-1}]$ is a UFD, then $R[X]$ is a UFD. This is a direct invocation of `nagata_theorem` with the submonoid S from Step 1. The one subtlety is that Mathlib's `IsLocalization.Away` structure on $R[X, X^{-1}]$ must be transported to a `UniqueFactorizationMonoid` instance on the abstract `Localization S`, via a ring equivalence and the fact that UFD transfers across multiplicative equivalences.

7.5 Step 5: the composite theorem

Finally, `polynomial_uniqueFactorizationMonoid_via_nagata` combines Steps 3 and 4: given a noetherian UFD R, it constructs the Laurent UFD instance and invokes Step 4 to obtain the polynomial UFD instance.

Remark 7.2. This result is also provable via the Gauss lemma (and is already available in Mathlib by that route). The file `Applications/Gauss.lean` wraps the Mathlib instance to make the comparison explicit. The value of the Nagata-based proof is not that it produces a new result, but that it exercises the formalized theorem in a setting where the prime-generated localization machinery is genuinely needed: the proof uses the full infrastructure (prime-generation, avoidance, transfer of irreducibility and primality) rather than reducing to a thin wrapper.

7.6 The primality transfer for Laurent polynomials

The most technically interesting ingredient in Step 3 is the primality transfer from $R[X]$ to $R[X, X^{-1}]$. The auxiliary lemma (formalized as `polynomial_toLaurent_prime_of_not_dvd_X`) states:

Lemma 7.3. *Let R be a UFD and let $r \in R[X]$ be a prime polynomial that does not divide X. Then the image of r in $R[X, X^{-1}]$ is prime.*

The proof uses the ideal-theoretic criterion for primality: it suffices to show that the image of the principal ideal $(r) \subseteq R[X]$ under the localization map is a prime ideal of $R[X, X^{-1}]$. Mathlib provides the result `IsLocalization.isPrime_of_isPrime_disjoint`, which states that the extension of a prime ideal to a localization remains prime whenever the ideal is disjoint from the multiplicative set. The condition $r \nmid X$ ensures precisely that (r) is disjoint from the powers of X: if $X^n \in (r)$ for some n, then $r \mid X^n$, so $r \mid X$ (since r is prime), contradicting the hypothesis.

This lemma illustrates a general pattern in localization-based arguments: the "interesting" primes in a localization are those that come from primes in the base ring whose principal ideals avoid the multiplicative set. The powers of X that are absorbed as units in $R[X, X^{-1}]$ are precisely the "uninteresting" primes; all other primes survive the localization.

7.7 Concrete instantiations

The file `Applications/Examples.lean` records several concrete consequences of the polynomial-ring theorem:

1. $\mathbb{Z}$ is a UFD (a direct wrapper around Mathlib's instance).
2. $\mathbb{Z}[X]$ is a UFD (combining the Mathlib integer UFD instance with the polynomial-ring theorem).
3. For any field k, the multivariate polynomial ring $k[X_1, \ldots, X_n]$ is a UFD (using Mathlib's `MvPolynomial.uniqueFactorizationMonoid`).
4. Iterated polynomial rings $R[X_1] \cdots [X_n]$ over a UFD R are UFDs.

These examples are deliberately thin—each is a one- or two-line invocation of existing Mathlib infrastructure—but they serve as machine-checkable witnesses that the formalization connects cleanly to the broader library ecosystem. In particular, the $\mathbb{Z}[X]$ example can be proved either via the Gauss lemma (Mathlib's default route) or via the Nagata theorem (our route through $\mathbb{Z}[X, X^{-1}]$), providing a concrete comparison point.

8 Lessons from the formalization

Several aspects of the formal development may be of interest to the proof-engineering community beyond the specific theorem.

8.1 Getting the hypothesis right

The most consequential design decision was the replacement of the prime-or-unit hypothesis with the prime-generated one. The prime-or-unit condition is superficially cleaner—it avoids quantifying over multisets—but it is essentially degenerate for submonoids with more than one non-unit prime generator, as discussed in Section 2.4.

The formalization made this defect visible earlier and more sharply than a textbook exposition would. The initial development compiled and type-checked under the prime-or-unit hypothesis; all transfer lemmas were proved correctly. The problem surfaced only at the application stage: the powers-of-X submonoid satisfies the prime-generated condition but *not* the prime-or-unit condition, since $X^2 \in S$ is neither prime nor a unit in $R[X]$. In a pen-and-paper development, one might paper over this gap by implicitly using the prime-generated formulation in the application while stating the theorem with the prime-or-unit hypothesis. The formal setting does not allow this.

The mathematical lesson is that a condition on individual elements of a submonoid is the wrong abstraction level when the argument requires factoring products of denominators; one must instead ask for a *factorization property* of the submonoid. This insight is not original—textbooks that treat the general case state the theorem with a prime-factorization hypothesis from the outset—but the formalization provided a concrete, machine-checkable failure mode.

8.2 Transfer lemmas as conditional infrastructure

The divisibility, irreducibility, and primality transfer lemmas are not unconditional equivalences between R and $S^{-1}R$. They hold under the specific combination of prime-generation and prime-avoidance hypotheses that arises in the Nagata argument. Formalizing them as self-contained lemmas (rather than inlining them into the main proof) required identifying exactly which hypotheses each transfer result needs.

For example, the divisibility lift (Lemma 4.3) requires both prime-generation of S and avoidance by p, while the primality transfer (Lemma 4.5) additionally requires that $p/1$ be prime in the localization. In the key lemma, the avoidance hypothesis is supplied by one branch of the case split, and the localization-side primality comes from the UFD hypothesis—but the transfer lemma itself does not know about either source. This decomposition is valuable not because each lemma is deep, but because the precise statement makes its reuse conditions explicit.

8.3 Multiset induction as the core combinatorial engine

The most labor-intensive part of the formalization is the multiset induction in the transfer lemmas. When clearing a denominator $s = q_1 \cdots q_n$ from a fraction, the argument peels off one prime factor at

a time, using primality of q_i to redirect the factor to the appropriate side of a product equation. This combinatorial bookkeeping—tracking which factors have been consumed, maintaining the inductive invariant, and handling the base case—is routine on paper but requires explicit management in the formal setting.

The auxiliary lemma `split_prime_factors_of_mul_eq` (which partitions a multiset of prime factors across the two sides of a product equation) is the longest single proof in the development. Its proof is a nested induction: the outer induction is on the multiset, and at each step the primality of the current factor determines which side of the partition absorbs it. The existence of this lemma as a self-contained, reusable result is a formalization artifact—on paper one would simply say "by induction on the number of prime factors"—but it clarifies the precise combinatorial content of the transfer-of-irreducibility argument.

8.4 Automation and transparency

Lean's tactic automation handles routine algebraic reasoning, but the key proof obligations—particularly the case analysis on irreducible elements dividing products of primes—require explicit combinatorial arguments beyond `simp` or `ring`. This is a feature, not a limitation: the formal proof preserves exactly the case distinctions that constitute the mathematical content.

The most effective automation patterns in this development were:

- `simp` with commutativity and associativity lemmas for rearranging products.
- `ac_rfl` for closing goals that differ only by associativity/commutativity.
- Lean's equation compiler for structural recursion on multisets (via `Multiset.induction_on`).

8.5 Localization ergonomics

Working with Mathlib's localization API involves navigating between the abstract `IsLocalization` typeclass and concrete `Localization` instances. The development stays close to the concrete type for definitional convenience, simplifying denominator manipulation at the cost of some generality.

A key design choice was the `dvd_of_iff` lemma, which characterizes divisibility of images in the localization:

$$\iota(a) \mid \iota(b) \iff \exists\, s \in S,\ a \mid s \cdot b.$$

This equivalence is the interface through which the Nagata transfer lemmas access the localization: they never manipulate fractions directly, but instead use it to translate localization-side divisibility into a ring-side equation amenable to multiset induction.

A future Mathlib-oriented refactoring might abstract over the localization construction, but for the present theorem the concrete approach proved practical.

8.6 Formalization effort and Lean 4 ergonomics

The total formalization comprises approximately 970 lines of Lean 4 source across 16 files, with 74 formally verified declarations (theorem/lemma/def) and zero uses of `sorry`, `admit`, or `native_decide`. The development was carried out over a period of several weeks by a single author using Lean 4.24.0 and Mathlib (commit hash recorded in `lean-toolchain` and `lake-manifest.json`).

Several Lean 4-specific ergonomic issues shaped the proof style:

- **Typeclass diamonds.** The `Localization` type carries multiple paths to `CommMonoidWithZero`, which occasionally forces the use of explicit `@` notation to disambiguate instances. The theorem statements in Section 5 show this pattern.

- **Multiset induction.** Key transfer lemmas reason by induction on the multiset of prime factors. Lean's `Multiset.induction_on` tactic handles the base and cons cases, but the recursive step often requires manual `calc` blocks to guide the simplifier.
- **Universe polymorphism.** All definitions and theorems are universe-polymorphic, which is necessary for eventual Mathlib compatibility but requires care in stating auxiliary lemmas to avoid universe metavariable errors.
- **Zero exclusion.** Several transfer lemmas require $0 \notin S$ as a hypothesis. In Lean 4, this is expressed as a `Fact` instance, which must be registered in the local context before typeclass inference can construct the localization's `IsDomain` instance. This boilerplate appears in the key lemma and in the main theorem's proof.

9 Related work

9.1 Mathematical background

The result first appeared in a short note by Nagata [11], which proved that a noetherian domain whose localization at a multiplicatively closed set of primes is a UFD is itself a UFD. Nagata subsequently included the theorem in his monograph on local rings [12]. Samuel's Tata lectures [14] give a particularly clean treatment of the single-prime-generator case and the polynomial-ring application. Matsumura's textbook [9] and Kaplansky's *Commutative Rings* [5] present the result in full generality for submonoids generated by primes, and our formulation follows this tradition.

The Stacks Project [15] provides a modern, self-contained exposition of the theorem (Tag 0AFU) in the language of commutative algebra, including a proof that any noetherian normal domain whose class group is trivial is a UFD. We followed the argument structure closest to Matsumura's and Samuel's accounts.

9.2 The Lean theorem prover and Mathlib

Our development is built in Lean 4 [10], a dependently typed functional programming language and interactive theorem prover. The mathematical foundation is provided by Mathlib [7], the community-maintained library of formalized mathematics for Lean 4.

Mathlib supplies all the algebraic infrastructure on which this project rests: the `Localization` type and `IsLocalization` typeclass; the `UniqueFactorizationMonoid` class and its characterizations; the `WfDvdMonoid` instance for noetherian domains; and the polynomial and Laurent polynomial rings with their ring-theoretic structure. Our contribution is the formal packaging of a Nagata-style transfer theorem and the proof architecture needed to make it applicable. The project depends on Mathlib but does not modify it.

9.3 Formal algebra in other proof assistants

In Coq, the Mathematical Components library [8] offers a mature algebraic environment with strong support for divisibility and polynomial algebra, including a Gauss-lemma-based proof that polynomial rings over UFDs are UFDs. The algebraic hierarchy is built on the SSReflect proof style, which favors boolean decision procedures over classical logic.

In Isabelle/HOL, Bordg [2] formalizes localization of commutative rings in the Archive of Formal Proofs, providing the ring-of-fractions construction and its universal property. Divasón et al. [4] formalize the Berlekamp–Zassenhaus factorization algorithm, demonstrating substantial factorization

infrastructure, though their focus is algorithmic factorization rather than structural factoriality theorems.

9.4 Related formalization projects in Lean

The closest thematic relative in Lean/Mathlib is the formalization of Dedekind domains and class groups by Baanen et al. [1], which also develops nontrivial algebraic infrastructure on top of Mathlib's localization and ideal theory. Their work includes formalized proofs about fractional ideals, the class group of a global field, and the finiteness of the class number, all of which require careful interaction with localization machinery similar to that needed for the Nagata theorem.

Among recent Annals of Formalized Mathematics publications, Brasca et al. [3] formalize Fermat's Last Theorem for regular primes in Lean 4, including a proof of Kummer's lemma via Hilbert's Theorems 90–94. Riou [13] formalizes derived categories and triangulated structures in Lean/Mathlib. Loeffler and Stoll [6] formalize the theory of zeta and L-functions, including a proof of Dirichlet's theorem on primes in arithmetic progressions. These projects share the methodological characteristic of building substantial mathematical results on top of Mathlib's existing infrastructure, and they illustrate the current state of the art for formal commutative algebra and number theory in Lean 4.

9.5 Novelty statement

No public formalization of Nagata's factoriality theorem is known to us in Lean/Mathlib, Coq/MathComp, or Isabelle/AFP. This is a statement about the public state of these libraries at the time of writing, not a claim that no unpublished or in-progress formalization exists.

10 Prospective library contributions

Not every result in the development is a candidate for upstreaming to Mathlib. We distinguish three tiers:

Good candidates.
The localization helpers and generic submonoid lemmas (e.g., `zero_notMem_of_primeGenerated`, `multiset_prod_mem_of_factors`) are reusable beyond the present theorem and fit naturally into Mathlib's localization and submonoid APIs.

Conditional candidates.
The `PrimeGenerated` predicate and its transfer lemmas are mathematically natural, but would need API stabilization and generalization to the `IsLocalization` typeclass before they fit Mathlib's abstraction boundaries.

Paper-level contribution.
The Nagata theorem itself and its specific proof packaging are the core contribution of this repository. Whether they belong in Mathlib depends on whether the library community considers the result sufficiently demanded to justify the maintenance cost.

11 Artifact and reproducibility

11.1 Artifact summary

The development targets Lean 4.24.0, builds with Lake (the Lean build system), and contains no `sorry`, `admit`, or `axiom` placeholders. Table 4 summarizes the artifact statistics.

Table 4: Artifact statistics.

Metric	Value
Lean source files	16
Lines of Lean source	970
Theorem/lemma declarations	74
`sorry/admit/axiom`	0
Lean version	4.24.0

11.2 Building the artifact

The artifact builds in two steps:

```
lake exe cache get     -- fetch prebuilt Mathlib oleans
lake build             -- build the project
```

The first command fetches prebuilt object files for the Mathlib dependency, avoiding a full rebuild of the library (which can take several hours). The second command compiles all project files, including the application layer. On a modern workstation with cached Mathlib, the project build completes in under two minutes.

11.3 Continuous integration

The repository includes a GitHub Actions workflow that runs `lake build` on every push and pull request, ensuring that the development remains buildable against its pinned Lean toolchain and Mathlib version.

11.4 Release packaging

The submission artifact is a tagged source release that freezes the toolchain pin, lock file, Lean sources, and this paper. Citation metadata is provided in a `CITATION.cff` file following the Citation File Format standard. The same release can be mirrored to a DOI-minting archive such as Zenodo if the venue accepts archival artifacts.

11.5 Verification and trust

The trusted base for the formalization consists of the Lean 4 kernel (which checks all proof terms), the Mathlib library (which provides algebraic infrastructure), and the Lake build system (which orchestrates compilation). No external oracles, custom axioms, or unverified code generation are used. The absence of `sorry` and `admit` can be mechanically verified by searching the source files.

12 Conclusion and future work

We have presented a Lean 4 formalization of Nagata's factoriality theorem under a prime-generated hypothesis on the multiplicative set, together with a worked application establishing unique factorization for polynomial rings via Laurent-polynomial localization. The formalization exercise surfaced a genuine mathematical issue—the degeneracy of the prime-or-unit hypothesis—and the resulting prime-generated formulation provides a reusable building block for localization-based factoriality arguments. The repository makes that reusability explicit by packaging the theorem not only in its

abstract prime-generated form, but also for submonoid closures generated by prime families and for the finite-prime-generator case.

Limitations. The formalization covers the multiplicative version of Nagata's theorem (for submonoids generated by primes). The more general ideal-theoretic formulations—such as the version involving t-operations or Krull domains—are not addressed. On the application side, the Laurent-polynomial example is intentionally compact; richer applications would further exercise the machinery. Some localization API choices are tailored to this project and may need refactoring for eventual Mathlib integration.

Future work. Several directions are natural continuations:

1. **Mathlib integration.** Abstracting the localization interface from the concrete `Localization` type to `IsLocalization` would make the results directly usable in Mathlib's ecosystem.
2. **Richer applications.** The formalized theorem could be applied to establish unique factorization in power-series rings, rings of algebraic integers, or other settings where a Nagata-style argument is natural.
3. **Generalization.** The ideal-theoretic generalization of Nagata's theorem (involving the class group and divisorial ideals) is a natural next target, though it would require substantially more infrastructure than is currently available in Mathlib.
4. **Comparison with Gauss-lemma approaches.** A formal comparison of the Nagata-based and Gauss-lemma-based proofs of the polynomial UFD theorem, both in proof complexity and dependency structure, would be an interesting study in formalization methodology.

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